

2.0.1 UGC NET CSE | July 2016 | Part 3 | Question: 75



A software program that infers and manipulates existing knowledge in order to generate new knowledge is known as:

- A. Data dictionary
C. Inference engine
B. Reference mechanism
D. Control strategy

ugcnetcse-july2016-paper3

Answer key

2.0.2 UGC NET CSE | July 2016 | Part 3 | Question: 66



A perceptron has input weights $W_1 = -3.9$ and $W_2 = 1.1$ with threshold value $T = 0.3$. What output does it give for the input $x_1 = 1.3$ and $x_2 = 2.2$?

- A. -2.65
B. -2.30
C. 0
D. 1

ugcnetcse-july2016-paper3

Answer key

2.1

Artificial Intelligence (17)

2.1.1 Artificial Intelligence: UGC NET CSE | December 2013 | Part 3 | Question: 26



The mean-end analysis process centers around the detection of differences between the current state and goal state. Once such a difference is isolated, an operator that can reduce the difference must be found. But perhaps that operator cannot be applied to the current state. So a sub-problem of getting to a state in which it can be applied is set up. The kind of backward chaining in which operators are selected and then sub goals are set up to establish the precondition of operators is called

- A. backward planning
C. operator subgoalting
B. goal stack planning
D. operator overloading

ugcnetcse-dec2013-paper3 artificial-intelligence

Answer key

2.1.2 Artificial Intelligence: UGC NET CSE | December 2014 | Part 3 | Question: 72



Match the following learning modes *w.r.t.* characteristics of available information for learning :

- | | |
|------------------|---|
| a. Supervised | i. Instructive information on desired responses, explicitly specified by a teacher. |
| b. Recording | ii. A priori design information for memory storing |
| c. Reinforcement | iii. Partial information about desired responses, or only "right" or "wrong" evaluative information |
| d. Unsupervised | iv. No information about desired responses |

Codes :

- | | a | b | c | d |
|----|----|-----|-----|----|
| A. | i | ii | iii | iv |
| B. | i | iii | ii | iv |
| C. | ii | iv | iii | i |
| D. | ii | iii | iv | i |

ugcnetcse-dec2014-paper3 artificial-intelligence machine-learning

Answer key

2.1.3 Artificial Intelligence: UGC NET CSE | December 2015 | Part 3 | Question: 46



Language model used in LISP is

- A. Functional programming
- B. Logic programming
- C. Object oriented programming
- D. All of the above

ugcnetcse-dec2015-paper3 artificial-intelligence

Answer key

2.1.4 Artificial Intelligence: UGC NET CSE | December 2015 | Part 3 | Question: 64



Consider the two class classification task that consists of the following points:

Class C_1 : $[-1, -1], [-1, 1], [1, -1]$

Class C_2 : $[1, 1]$

The decision boundary between the two classes C_1 and C_2 using single perception is given by:

- A. $x_1 - x_2 - 0.5 = 0$
- B. $-x_1 - x_2 - 0.5 = 0$
- C. $0.5(x_1 + x_2) - 1.5 = 0$
- D. $x_1 + x_2 - 0.5 = 0$

ugcnetcse-dec2015-paper3 artificial-intelligence

Answer key

2.1.5 Artificial Intelligence: UGC NET CSE | January 2017 | Part 3 | Question: 55



Consider following two rules R1 and R2 in logical reasoning in Artificial Intelligence (AI):

R1 : From $\alpha \supset \beta$ $\frac{\text{and } \alpha}{\text{Inter } \beta}$ is known as Modulus Tollens (MT)

R2 : From $\alpha \supset \beta$ $\frac{\text{and } \neg \beta}{\text{Inter } \neg \alpha}$ is known as Modus Ponens(MP)

- A. Only R1 is correct.
- B. Only R2 is correct.
- C. Both R1 and R2 are correct.
- D. Neither R1 nor R2 is correct.

ugcnetcse-jan2017-paper3 non-gatecse artificial-intelligence

Answer key

2.1.6 Artificial Intelligence: UGC NET CSE | July 2018 | Part 2 | Question: 71



In artificial Intelligence (AI), an environment is uncertain if it is ____

- A. Not fully observable and not deterministic
- B. Not fully observable or not deterministic
- C. Fully observable but not deterministic
- D. Not fully observable but deterministic

ugcnetcse-july2018-paper2 artificial-intelligence

Answer key

2.1.7 Artificial Intelligence: UGC NET CSE | July 2018 | Part 2 | Question: 72



In artificial Intelligence (AI), a simple reflex agent selects actions on the basis of ____

- A. current percept, completely ignoring rest of the percept history
- B. rest of the percept history, completely ignoring the current percept
- C. both current percept and complete percept history
- D. both current percept and just previous percept

ugcnetcse-july2018-paper2 artificial-intelligence

Answer key

2.1.8 Artificial Intelligence: UGC NET CSE | July 2018 | Part 2 | Question: 73



In heuristic search algorithms in Artificial Intelligence (AI), if a collection of admissible heuristics $h_1 \dots h_m$ is available for a problem and none of them dominates any of the others, which should we choose?

- A. $h(n) = \max\{h_1(n), \dots, h_m(n)\}$
B. $h(n) = \min\{h_1(n), \dots, h_m(n)\}$
C. $h(n) = \text{avg}\{h_1(n), \dots, h_m(n)\}$
D. $h(n) = \text{sum}\{h_1(n), \dots, h_m(n)\}$

ugcnetcse-july2018-paper2 artificial-intelligence

Answer key

2.1.9 Artificial Intelligence: UGC NET CSE | July 2018 | Part 2 | Question: 74



Consider following sentences regarding A^* , an informed search strategy in Artificial Intelligence (AI).

- a. A^* expands all nodes with $f(n) < C^*$
b. A^* expands no nodes with $f(n) \geq C^*$
c. Pruning is integral to A^*

Here, C^* is the cost of the optimal solution path. Which of the following is correct with respect to the above statements?

- A. Both statements a and statement b are true
B. Both statements a and statement c are true
C. Both statements b and statement c are true
D. All the statements a, b and c are true

ugcnetcse-july2018-paper2 artificial-intelligence

Answer key

2.1.10 Artificial Intelligence: UGC NET CSE | June 2012 | Part 3 | Question: 2



In Delta Rule for error minimization

- A. weights are adjusted w.r.to change in the output
B. weights are adjusted w.r.to difference between desired output and actual output
C. weights are adjusted w.r.to difference between output and output
D. none of the above

ugcnetcse-june2012-paper3 artificial-intelligence machine-learning

Answer key

2.1.11 Artificial Intelligence: UGC NET CSE | June 2012 | Part 3 | Question: 21



A^* algorithm uses $f' = g + h'$ to estimate the cost of getting from the initial state to the goal state, where g is a measure of cost getting from initial state to the current node and the function h' is an estimate of the cost of getting from the current node to the goal state. To find a path involving the fewest number of steps, we should test,

- A. $g = 1$ B. $g = 0$ C. $h' = 0$ D. $h' = 1$

ugcnetcse-june2012-paper3 artificial-intelligence

Answer key

2.1.12 Artificial Intelligence: UGC NET CSE | June 2014 | Part 3 | Question: 28



Match the following :

List – I

List – II

- | | |
|--------------------------------|--------------------------|
| a. Expert systems | i. Pragmatics |
| b. Planning | ii. Resolution |
| c. Prolog | iii. means-end analysis |
| d. Natural language processing | iv. Explanation facility |

Codes :

- A. a-iii, b-iv, c-i, d-ii
C. a-i, b-ii, c-iii, d-iv

- B. a-iii, b-iv, c-ii, d-i
D. a-iv, b-iii, c-ii, d-i

ugcnetjune2014iii artificial-intelligence

Answer key 

2.1.13 Artificial Intelligence: UGC NET CSE | June 2014 | Part 3 | Question: 30



Slots and facets are used in

- A. Semantic Networks B. Frames C. Rules D. All of these

ugcnetjune2014iii artificial-intelligence

Answer key 

2.1.14 Artificial Intelligence: UGC NET CSE | Junet 2015 | Part 3 | Question: 23



Which of the following is false for the programming language PROLOG?

- A. A PROLOG variable can only be assigned to a value once
B. PROLOG is a strongly typed language
C. The scope of a variable in PROLOG is a single clause or rule
D. The scope of a variable in PROLOG is a single query

ugcnetcse-june2015-paper3 artificial-intelligence

Answer key 

2.1.15 Artificial Intelligence: UGC NET CSE | Junet 2015 | Part 3 | Question: 56



Match the following knowledge representation techniques with their applications :

List – I

List – II

- | | |
|-----------------------------|---|
| (a) Frames | (i) Pictorial representation of objects, their attributes and relationships |
| (b) Conceptual dependencies | (ii) To describe real world stereotype events |
| (c) Associative networks | (iii) Record like structures for grouping closely related knowledge |
| (d) Scripts | (iv) Structures and primitives to represent sentences |

Codes :

- A. (a)-(ii), (b)-(iv), (c)-(i), (d)-(ii)
C. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

- B. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

ugcnetcse-june2015-paper3 artificial-intelligence

Answer key 

2.1.16 Artificial Intelligence: UGC NET CSE | Junet 2015 | Part 3 | Question: 59



Match the following with respect to heuristic search techniques :

List – I

- (a) Steepest-accent Hill Climbing
- (b) Branch-and-bound
- (c) Constraint satisfaction
- (d) Means-end-analysis

List – II

- (i) Keeps track of all partial paths which can be candidate for further explanation
- (ii) Discover problem state(s) that satisfy a set of constraints
- (iii) Detects difference between current state and goal state
- (iv) Considers all moves from current state and selects best move

Codes :

- A. (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- C. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

- B. (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- D. (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

ugcnetcse-june2015-paper3 artificial-intelligence

Answer key **2.1.17 Artificial Intelligence: UGC NET CSE | October 2020 | Part 2 | Question: 36**

Which of the following is NOT true in problem solving in artificial intelligence?

- A. Implements heuristic search technique
- B. Solution steps are not explicit
- C. Knowledge is imprecise
- D. It works on or implements repetition mechanism

ugcnetcse-oct2020-paper2 non-gatecse artificial-intelligence

Answer key **2.2****Artificial Neural Network (1)****2.2.1 Artificial Neural Network: UGC NET CSE | December 2013 | Part 3 | Question: 30**

An artificial neuron receives n inputs $x_1, x_2, \dots, x_n$ with weights $w_1, w_2, \dots, w_n$ attached to the input links. The weighted sum _____ is computed to be passed on to a non-linear filter ϕ called activation function to release the output.

- A. $\sum w_i$
- B. $\sum x_i$
- C. $\sum w_i + \sum x_i$
- D. $\sum w_i \cdot \sum x_i$

ugcnetcse-dec2013-paper3 machine-learning artificial-neural-network

Answer key **2.3****Back Propagation (1)****2.3.1 Back Propagation: UGC NET CSE | December 2012 | Part 2 | Question: 46**

Back propagation is a learning technique that adjusts weights in the neural network by propagating weight changes.

- A. Forward from source to sink
- B. Backward from sink to source
- C. Forward from source to hidden nodes
- D. Backward from sink to hidden nodes

ugcnetcse-dec2012-paper2 machine-learning data-mining back-propagation

Answer key **2.4****Blocks World Problem (1)****2.4.1 Blocks World Problem: UGC NET CSE | September 2013 | Part 3 | Question: 5**

The Blocks World Problem in Artificial Intelligence is normally discussed to explain a

- A. Search technique
- B. Planning system
- C. Constraint satisfaction system
- D. Knowledge base system

ugcnetcse-sep2013-paper3 artificial-intelligence blocks-world-problem

Answer key 

2.5 Chaining (1)

2.5.1 Chaining: UGC NET CSE | December 2015 | Part 3 | Question: 8



Forward chaining systems are _____ where as backward chaining systems are _____

- A. Data driven, Data driven
- B. Goal driven, Data driven
- C. Data driven, Goal driven
- D. Goal driven, Goal driven

ugcnetcse-dec2015-paper3 artificial-intelligence chaining

Answer key 

2.6 Expert System (2)

2.6.1 Expert System: UGC NET CSE | December 2012 | Part 3 | Question: 67



An expert system shell is an expert system without

- A. domain knowledge
- B. explanation facility
- C. reasoning with knowledge
- D. all of the above

ugcnetcse-dec2012-paper3 artificial-intelligence expert-system

Answer key 

2.6.2 Expert System: UGC NET CSE | December 2015 | Part 3 | Question: 45



Reasoning strategies used in expert systems include

- A. Forward chaining, backward chaining and problem reduction
- B. Forward chaining, backward chaining and boundary mutation
- C. Forward chaining, backward chaining and back propagation
- D. Forward chaining, problem reduction and boundary mutation

ugcnetcse-dec2015-paper3 artificial-intelligence expert-system

Answer key 

2.7 Fuzzy Logic (1)

2.7.1 Fuzzy Logic: UGC NET CSE | June 2019 | Part 2 | Question: 94



A fuzzy conjunction operator denoted as $t(x, y)$ and a fuzzy disjunction operator denoted as $s(x, y)$ form a dual pair if they satisfy the condition:

- A. $t(x, y) = 1 - s(x, y)$
- B. $t(x, y) = s(1 - x, 1 - y)$
- C. $t(x, y) = 1 - s(1 - x, 1 - y)$
- D. $t(x, y) = s(1 + x, 1 + y)$

ugcnetcse-june2019-paper2 artificial-intelligence fuzzy-logic

Answer key 

2.8 Genetic Algorithms (1)

2.8.1 Genetic Algorithms: UGC NET CSE | June 2019 | Part 2 | Question: 97



Consider the following:

- i. Evolution
- ii. Selection
- iii. Reproduction
- iv. Mutation

Which of the following are found in genetic algorithms?

A. b, c and d only

B. b and d only

C. a, b, c and d

D. a, b and d only

ugcnetcse-june2019-paper2 artificial-intelligence genetic-algorithms

Answer key 

2.9

Heuristic Search (1)

2.9.1 Heuristic Search: UGC NET CSE | December 2013 | Part 3 | Question: 25



If h^* represents an estimate from the cost of getting from the current node N to the goal node and h represents actual cost of getting from the current node to the goal node, then A^* algorithm gives an optimal solution if

A. h^* is equal to h

B. h^* overestimates h

C. h^* underestimates h

D. none of these

ugcnetcse-dec2013-paper3 artificial-intelligence heuristic-search

Answer key 

2.10

Linear Programming (1)

2.10.1 Linear Programming: UGC NET CSE | September 2013 | Part 3 | Question: 13



If an artificial variable is present in the 'basic variable' of optimal simplex table then the solution is

A. Alternative solution

B. Infeasible solution

C. Unbounded solution

D. Degenerate solution

ugcnetcse-sep2013-paper3 artificial-intelligence linear-programming

Answer key 

2.11

Machine Learning (2)

2.11.1 Machine Learning: UGC NET CSE | December 2012 | Part 3 | Question: 73



Match the following:

a. Supervised learning 1. The decision system receives rewards for its action at the end of a sequence of steps

b. Unsupervised learning 2. Manual labels of inputs are not used

c. Reinforcement learning 3. Manual labels of inputs are used

d. Inductive learning 4. System learns by example

a b c d

A 1 2 3 4

B 2 3 1 4

C 3 2 4 1

D 3 2 1 4

ugcnetcse-dec2012-paper3 machine-learning

Answer key 

2.11.2 Machine Learning: UGC NET CSE | June 2014 | Part 3 | Question: 09



Perceptron learning, Delta learning and *LMS* learning are learning methods which falls under the category of

A. Error correction learning - learning with a teacher

B. Reinforcement learning - learning with a critic

- C. Hebbian learning
- D. Competitive learning - learning without a teacher

ugcnetjune2014iii machine-learning

Answer key 

2.12 Map Coloring (1)

2.12.1 Map Coloring: UGC NET CSE | June 2013 | Part 3 | Question: 71



The map colouring problem can be solved using which of the following technique?

- A. Means-end analysis
- B. Constraint satisfaction
- C. AO* search
- D. Breadth first search

ugcnetcse-june2013-paper3 artificial-intelligence map-coloring

Answer key 

2.13 Means End (1)

2.13.1 Means End: UGC NET CSE | September 2013 | Part 3 | Question: 6



Means-Ends Analysis process centres around the detection of difference between the current state and the goal state. Once such a difference is found, then to reduce the difference one applies

- A. a forward search that can reduce the difference
- B. a backward search that can reduce the difference
- C. a bidirectional search that can reduce the difference
- D. an operator that can reduce the difference

ugcnetcse-sep2013-paper3 artificial-intelligence means-end analysis

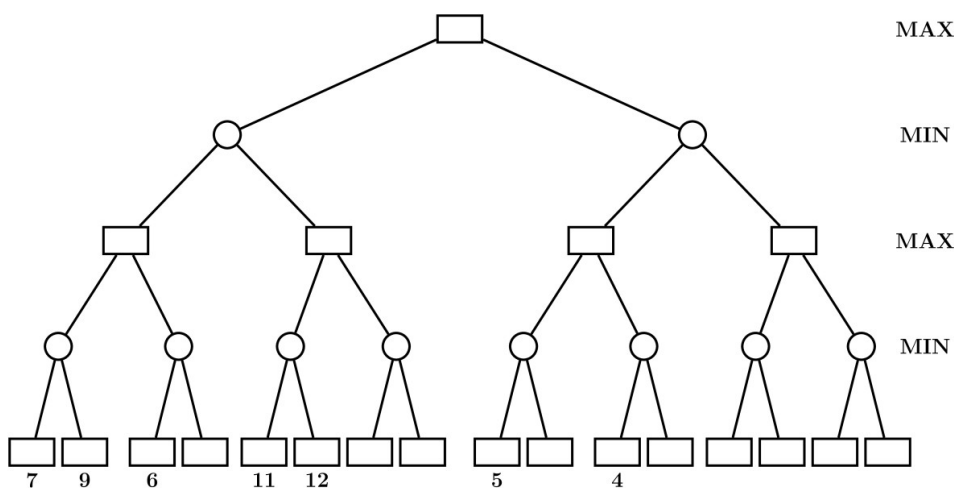
Answer key 

2.14 Minimax Procedure (1)

2.14.1 Minimax Procedure: UGC NET CSE | June 2019 | Part 2 | Question: 91



Consider the game tree given below:



Here ○ and □ represents MIN and MAX nodes respectively. The value of the root node of the game tree is

- A. 4
- B. 7
- C. 11
- D. 12

ugcnetcse-june2019-paper2 artificial-intelligence minimax-procedure

Answer key 

2.15

Neural Network (3)

2.15.1 Neural Network: UGC NET CSE | December 2012 | Part 3 | Question: 9



You are given an OR problem and XOR problem to solve. Then, which one of the following statements is true?

- A. Both OR and XOR problems can be solved using single layer perception
- B. OR can be solved using single layer perception and XOR problem can be solved using self organizing maps
- C. OR problem can be solved using radial basis function and XOR problem can be solved using single layer perception
- D. OR can be solved using single layer perception and XOR problem can be solved using radial basis function

ugcnetcse-dec2012-paper3 artificial-intelligence neural-network

Answer key

2.15.2 Neural Network: UGC NET CSE | June 2019 | Part 2 | Question: 98



Which of the following is an example of unsupervised neural network?

- A. Back-propagation network
- B. Hebb network
- C. Associative memory network
- D. Self-organizing feature map

ugcnetcse-june2019-paper2 artificial-intelligence neural-network

Answer key

2.15.3 Neural Network: UGC NET CSE | September 2013 | Part 3 | Question: 28



In a single perceptron, the updation rule of weight vector is given by

- A. $w(n+1) = w(n) + \eta[d(n) - y(n)]$
- B. $w(n+1) = w(n) - \eta[d(n) - y(n)]$
- C. $w(n+1) = w(n) + \eta[d(n) - y(n)] * x(n)$
- D. $w(n+1) = w(n) - \eta[d(n) - y(n)] * x(n)$

ugcnetcse-sep2013-paper3 neural-network machine-learning

Answer key

2.16

Planning (1)

2.16.1 Planning: UGC NET CSE | July 2018 | Part 2 | Question: 78



Consider the following two sentences:

- a. The planning graph data structure can be used to give a better heuristic for a planning problem
- b. Dropping negative effects from every action schema in a planning problem results in a relaxed problem

Which of the following is correct with respect to the above sentences?

- A. Both sentence a and sentence b are false
- B. Both sentence a and sentence b are true
- C. Sentence a is true but sentence b is false
- D. Sentence a is false but sentence b is true

ugcnetcse-july2018-paper2 planning

Answer key

2.17

Prolog (2)

2.17.1 Prolog: UGC NET CSE | June 2013 | Part 3 | Question: 67



Which one of the following is the correct implementation of the meta-predicate “not” in PROLOG (Here G represents a goal)?

- A. `not(G):- !, call(G), fail. not(G).`
- B. `not(G):- call(G), !, fail. not(G).`
- C. `not(G):- call(G), fail, !, not(G).`
- D. `not(G):- call(G), !, fail.not(G):- !.`

2.17.2 Prolog: UGC NET CSE | June 2014 | Part 3 | Question: 14

Which one of the following describes the syntax of prolog program?

- I. Rules and facts are terminated by full stop(.)
- II. Rules and facts are terminated by semi colon(;)
 - III. Variables names must start with upper case alphabets.
 - IV. Variables names must start with lower case alphabets.

- A. I, II B. III, IV C. I, III D. II, IV

ugcnetjune2014iii artificial-intelligence prolog

Answer key

2.18 Reinforcement Learning (1)**2.18.1 Reinforcement Learning: UGC NET CSE | June 2019 | Part 2 | Question: 100**

Reinforcement learning can be formalized in terms of _____ in which the agent initially only knows the set of possible _____ and the set of possible actions.

- A. Markov decision processes, objects B. Hidden states, objects
C. Markov decision processes, states D. objects, states

ugcnetcse-june2019-paper2 artificial-intelligence reinforcement-learning

Answer key

2.19 Searches (1)**2.19.1 Searches: UGC NET CSE | June 2019 | Part 2 | Question: 92**

Math List-I with List-II:

	List-I		List-II
(a)	Greedy best-first	(i)	Minimal cost $(p) + h(p)$
(b)	Lowest cost-first	(ii)	Minimal $h(p)$
(c)	A^* algorithm	(iii)	Minimal cost (p)

Choose the correct option from those given below:

- A. (a) – (i) ; (b) – (ii); (c) – (iii)
B. (a) – (iii) ; (b) – (ii); (c) – (i)
C. (a) – (i) ; (b) – (iii); (c) – (ii)
D. (a) – (ii) ; (b) – (iii); (c) – (i)

ugcnetcse-june2019-paper2 artificial-intelligence searches

Answer key

2.20 Sigmoid Function (1)**2.20.1 Sigmoid Function: UGC NET CSE | June 2019 | Part 2 | Question: 99**

The value of the derivative of Sigmoid function given by $f(x) = \frac{1}{1 + e^{-2x}}$ at $x = 0$ is

- A. 0 B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. ∞

ugcnetcse-june2019-paper2 artificial-intelligence sigmoid-function

Answer key

2.21**Strips (1)****2.21.1 Strips: UGC NET CSE | June 2019 | Part 2 | Question: 93**

The STRIPS representation is

- A. a feature-centric representation
- B. an action-centric representation
- C. a combination of feature-centric and action-centric representations
- D. a hierarchical feature-centric representation

ugcnetcse-june2019-paper2 artificial-intelligence strips

Answer key

2.22**User Query (1)****2.22.1 User Query: Required Artificial Intelligence resources for UGC NET**

Hi

I need some useful resources of AI for upcoming UGC NET exam.

Currently, I am reading from Rich and Knight, Is it enough?

Please let me know about any good quality books or video lectures.

user-query

Answer key

Answer Keys

2.0.1	C	2.0.2	C	2.1.1	C	2.1.2	A	2.1.3	C
2.1.4	B	2.1.5	D	2.1.6	B	2.1.7	Q-Q	2.1.8	Q-Q
2.1.9	Q-Q	2.1.10	B	2.1.11	A	2.1.12	D	2.1.13	B
2.1.14	B	2.1.15	A	2.1.16	B	2.1.17	D	2.2.1	D
2.3.1	B	2.4.1	B	2.5.1	D	2.6.1	A	2.6.2	B
2.7.1	C	2.8.1	C	2.9.1	C	2.10.1	B	2.11.1	D
2.11.2	A	2.12.1	B	2.13.1	D	2.14.1	B	2.15.1	D
2.15.2	D	2.15.3	C	2.16.1	Q-Q	2.17.1	B	2.17.2	C
2.18.1	C	2.19.1	D	2.20.1	B	2.21.1	B	2.22.1	Q-Q